

Organization Sample Network

As a part of your end year networking project, you are required to design and implement Vic Modern Hotel network. The hotel has three floors; in the first floor there three departments (Reception, store and Logistics), in the second floor there are three departments (Finance, HR and Sales/Marketing), while the third floor hosts the IT and Admin. Therefore, the following are part of the considerations during the design and implementation.

1. There should be three routers connecting each floor (all placed in the server room in IT department).
2. All routers should be connected to each other using serial DCE cable.
3. The network between the routers should be 10.10.10.0/30, 10.10.10.4/30, 10.10.10.8/30
4. Each floor is expected to have one switch (placed in the respective floor).
5. Each floor is expected to have WIFI networks connected to laptops and phones.
6. Each department is expected to have a printer.
7. Each department is expected to be in different VLAN with the following details

1st Floor;

- Reception- VLAN 80, Network of 192.168.8.0/24
- Store- VLAN 70, Network of 192.168.7.0/24
- Logistics- VLAN 60, Network of 192.168.6.0/24

2nd Floor;

- Finance- VLAN 50, Network of 192.168.5.0/24
- HR- VLAN 40, Network of 192.168.4.0/24
- Sales- VLAN 30, Network of 192.168.3.0/24

3rd Floor;

- Admin- VLAN 20, Network of 192.168.2.0/24
- IT- VLAN 10, Network of 192.168.1.0/24

8. Use OSPF as the routing protocol to advertise routes.
9. All devices in the network are expected to obtain IP address dynamically with their respective router configured as the DHCP server.
10. All the devices in the network are expected to communicate with each other.
11. Configure SSH in all the routers for remote login.
12. In IT department, add PC called Test-PC to port fa0/1 and use it to test remote login.
13. Configure port security to IT-dept switch to allow only Test-PC to access port fa0/1 (use sticky method to obtain mac-address with violation mode of shutdown.)

Floor-3 Router

```
Router(config-if)#int se0/3/1
Router(config-if)#clock rate 64000
Router(config-if)#do wr
Router(config-if)#int se0/3/0
Router(config-if)#clock rate 64000
Router(config-if)#do wr
Router(config)#int se0/3/1
Router(config-if)#ip address 10.10.10.6 255.255.255.252
Router(config-if)#int se0/3/0
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#do wr
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.4 255.255.255.252 area 0
Router(config-router)#network 10.10.10.0 255.255.255.252 area 0
Router(config-router)#network 192.168.2.0 255.255.255.0 area 0
Router(config-router)#network 192.168.1.0 255.255.255.0 area 0
Router(config-router)#do wr
Router(config)#int gig0/0.20
Router(config-subif)#encapsulation dot1Q 20
```

```
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int gig0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.1 255.255.255.0
Router(config-subif)#exit
Router(config)#service dhcp
Router(config)#ip dhcp pool Admin
Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.2.1
Router(dhcp-config)#dns-server 192.168.2.1
Router(dhcp-config)#exit
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 192.168.1.1
Router(dhcp-config)#exit
Router(config)#hostname Floor3
Floor3(config)#ip domain-name Third
Floor3(config)#username third pass third
Floor3(config)#crypto key generate rsa
Floor3(config)#line vty 0 15
Floor3(config-line)#login local
Floor3(config-line)#transport input ssh
Floor3(config-line)#do wr
```

Floor-3 Switch

```
Switch(config)#int range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config)#int range fa0/4-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#do wr
Switch(config-if-range)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#do wr
Switch(config)#int fa0/5
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address sticky
```

```
Switch(config-if)#switchport port-security violation shutdown
```

```
Switch(config-if)#do wr
```

Floor-2 Router

```
Router(config)#int se0/3/1
```

```
Router(config-if)#no sh
```

```
Router(config)#int se0/3/0
```

```
Router(config-if)#no sh
```

```
Router(config)#int se0/3/1
```

```
Router(config-if)#ip address 10.10.10.10 255.255.255.252
```

```
Router(config-if)#int se0/3/0
```

```
Router(config-if)#ip address 10.10.10.1 255.255.255.252
```

```
Router(config-if)#do wr
```

```
Router(config)#int gig0/0.50
```

```
Router(config-subif)#encapsulation dot1Q 50
```

```
Router(config-subif)#ip address 192.168.5.1 255.255.255.0
```

```
Router(config-subif)#exit
```

```
Router(config)#int gig0/0.40
```

```
Router(config-subif)#encapsulation dot1Q 40
```

```
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
```

```
Router(config-subif)#exit
```

```
Router(config)#int gig0/0.30
```

```
Router(config-subif)#encapsulation dot1Q 30
```

```
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
```

```
Router(config-subif)#exit
```

```
Router(config)#service dhcp
```

```
Router(config)#ip dhcp pool Finance
```

```
Router(dhcp-config)#network 192.168.5.0 255.255.255.0
```

```
Router(dhcp-config)#default-router 192.168.5.1
```

```
Router(dhcp-config)#dns-server 192.168.5.1
```

```
Router(dhcp-config)#exit
```

```
Router(config)#ip dhcp pool HR
```

```
Router(dhcp-config)#network 192.168.4.0 255.255.255.0
```

```
Router(dhcp-config)#default-router 192.168.4.1
```

```
Router(dhcp-config)#dns-server 192.168.4.1
```

```
Router(dhcp-config)#exit
```

```
Router(config)#ip dhcp pool Sales
```

```
Router(dhcp-config)#network 192.168.3.0 255.255.255.0
```

```
Router(dhcp-config)#default-router 192.168.3.1
```

```
Router(dhcp-config)#dns-server 192.168.3.1
```

```
Router(dhcp-config)#exit
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.8 255.255.255.252 area 0
Router(config-router)#network 10.10.10.0 255.255.255.252 area 0
Router(config-router)#network 192.168.5.0 255.255.255.0 area 0
Router(config-router)#network 192.168.4.0 255.255.255.0 area 0
Router(config-router)#network 192.168.3.0 255.255.255.0 area 0
Router(config-router)#do wr
Router(config)#hostname Floor2
Floor2(config)#ip domain-name Second
Floor2(config)#username second pass second
Floor2(config)#crypto key generate rsa
Floor2(config)#line vty 0 15
Floor2(config-line)#login local
Floor2(config-line)#transport input ssh
Floor2(config-line)#do wr
```

Floor-2 Switch

```
Switch(config)#int range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 50
Switch(config-if-range)#int range fa0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#int range fa0/6-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#do wr
Switch(config-if-range)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#do wr
```

Floor-1 Router

```
Router(config)#int se0/3/0
Router(config-if)#no sh
Router(config)#int se0/3/1
Router(config-if)#no sh
Router(config)#int se0/3/1
Router(config-if)#ip address 10.10.10.5 255.255.255.252
Router(config-if)#int se0/3/0
```

```
Router(config-if)#ip address 10.10.10.9 255.255.255.252
Router(config)#int gig0/0.80
Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip address 192.168.8.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int gig0/0.70
Router(config-subif)#encapsulation dot1Q 70
Router(config-subif)#ip address 192.168.7.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int gig0/0.80
Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip address 192.168.8.1 255.255.255.0
Router(config-subif)#exit
Router(config)#service dhcp
Router(config)#ip dhcp pool Reception
Router(dhcp-config)#network 192.168.8.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.8.1
Router(dhcp-config)#dns-server 192.168.8.1
Router(dhcp-config)#exit
Router(config)#ip dhcp pool Store
Router(dhcp-config)#network 192.168.7.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.7.1
Router(dhcp-config)#dns-server 192.168.7.1
Router(dhcp-config)#exit
Router(config)#ip dhcp pool Logistics
Router(dhcp-config)#network 192.168.6.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.6.1
Router(dhcp-config)#dns-server 192.168.6.1
Router(dhcp-config)#exit
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.4 255.255.255.252 area 0
Router(config-router)#network 10.10.10.8 255.255.255.252 area 0
Router(config-router)#network 192.168.8.0 255.255.255.0 area 0
Router(config-router)#network 192.168.7.0 255.255.255.0 area 0
Router(config-router)#network 192.168.6.0 255.255.255.0 area 0
Router(config-router)#do wr
Router(config)#hostname Floor1
Floor1(config)#ip domain-name First
Floor1(config)#username first pass first
Floor1(config)#crypto key generate rsa
```

```
Floor1(config)#line vty 0 15
Floor1(config-line)#login local
Floor1(config-line)#transport input ssh
Floor1(config-line)#do wr
```

Floor-1 Switch

```
Switch(config)#int range fa0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 80
Switch(config-if-range)#int range fa0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 70
Switch(config-if-range)#int range fa0/6-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 60
Switch(config-if-range)#do wr
Switch(config-if-range)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#do wr
```